

# Jeffrey Hoelzel Jr

[jeffreyhoelzeljr@gmail.com](mailto:jeffreyhoelzeljr@gmail.com) | [linkedin.com/in/jeffrey-hoelzel-jr](https://www.linkedin.com/in/jeffrey-hoelzel-jr) | [github.com/jeffreyHoelzel](https://github.com/jeffreyHoelzel) | [jeffreyhoelzel.com](https://jeffreyhoelzel.com)

## Work Experience

### Machine Learning Engineer (Research) | Pathogen and Microbiome Institute | Flagstaff, AZ Aug 2025 - Present

- Developed PepSeqPred, a **PyTorch**-based **machine learning (ML)** pipeline to predict antibody epitope locations across diverse pathogens.
- Optimized **ESM-2** embedding generation pipelines using batching and **SLURM GPU parallelization**, reducing high-performance compute (HPC) runtime by over **50%**.
- Built **logging** and **evaluation** tools to validate embeddings and model outputs for downstream training and predictions.

### Software Engineer, Data Systems | NAU SICCS | Flagstaff, AZ Jan 2025 - Present

- Engineered **Python**-based **web scrapers** with **Selenium** and **BeautifulSoup** to extract unstructured UltraSignUp trail race data for downstream statistical analysis.
- Improved scraping throughput by nearly **90%** via **concurrency** and **adaptive wait time** strategies.
- Increased reliability by implementing dynamic **retry logic**, **exponential backoff**, and centralized **logging**.
- Reduced OpenStreetMap API calls by **48.5%** through dynamic GPX coordinate batching.

### Information Technology Intern | Cavco Industries, Inc. | Phoenix, AZ Jun 2025 - Aug 2025

- Built agentic **large-language model (LLM)** workflows, enabling automated ServiceDesk Plus ticket operations.
- Designed a **retrieval-augmented generation (RAG)** pipeline, reducing response latency by **60%** and consolidating **3+ API** calls into **1** request.
- Developed an ML pipeline clustering **100,000+** tickets to automatically generate Confluence knowledge articles.
- Implemented **persistent session** and **context** handling for multi-turn chatbot interactions.

### Software Engineer, Applied AI | CANIS Lab | Flagstaff, AZ Jan 2025 - Jul 2025

- Built a campus health chatbot using **ReactJS** and **Flask** to support student health inquiries and facilitate research studies.
- Evaluated multiple language models on **multi-GPU cloud compute** systems, enabling **sub-second** response times.
- Improved response relevance by **35%** by refining **RAG** pipelines and **contextual ranking**.

## Projects

### ArtemiS3 | Capstone Project (USGS and NASA) Svelte | FastAPI | PostgreSQL | Boto3 | Meilisearch

- Architected a **Dockerized** search platform enabling discovery across public NASA and USGS **AWS S3** datasets.
- Built **REST API** endpoints with **FastAPI** and optimized queries using **Meilisearch** and **Boto3**.
- Led technical planning and stakeholder presentations, delivering **architecture**, **feasibility**, and **requirements** documentation.

### MACH-IV Clustering | Unsupervised Machine Learning (CS472) Scikit-Learn | SciPy | Pandas | NumPy

- Designed and implemented an end-to-end **clustering** pipeline analyzing **70,000+** participant responses.
- Applied **k-means**, **Gaussian mixture**, **hierarchical**, and **spectral clustering** to identify latent behavioral patterns.
- Architected the project codebase and deployed an **interactive results site** for model exploration.

### Louie's Ratings | Software Engineering (CS386) ReactJS | Flask | PostgreSQL

- Led a five-person team through sprint planning and delivery of a **full-stack** academic rating and grade insights platform.
- Developed **secure authentication** flows using **ReactJS** and **Flask** with **bcrypt** password hashing.
- Implemented **unit** and end-to-end **acceptance** testing with **unittest** and **Selenium** to ensure deployment stability.

## Education

### Northern Arizona University, Flagstaff, AZ GPA: 3.85

BSc in Software Engineering | Minor in Mathematics

Lumberjack Scholar | HURA Grant Recipient | Dean's List

## Technical Skills

**Languages:** Python, JavaScript, TypeScript, C, Java, SQL, HTML, CSS

**Libraries/Frameworks:** ReactJS, Svelte, Flask, FastAPI, PyTorch, Scikit-Learn, Pandas, SQLite, PostgreSQL, Firebase Firestore

**Developer Tools:** Docker, Git, Slurm (Bash), AWS, Google Cloud, HPC, CI/CD, Agile